

06-02-2026	cross validation & clustering	k-fold cv -- when the models are overfitting there we implement k fold cv dataset we divide into 1000 records divided into 5 fold . Each folds contain 200 recording build model 5 times average of all the 5 models then we calculate (reduce high var - low var) reduce overfitting technique model tuning -- gridsearch cv & random search cv we pass each and every model parameters (manually assigne all the parameter in one go) grid search cv will finetune optimized parameter discussed math for k-meas clustering -- grouping no dependent variable data is discrete we are using customer segmentation, rating the food, grading score student elbow method -- will decide how to make cluster from the dataset xaxis - number of cluster & y axis - wcss (within cluster sum of sward)
07-02-2026	k-means & hierarchical clustering & dbscan	kmeans preacticle we worked, hierachical cluster -- 2 part agglomerative clustering -- using dendogram we build the cluster from bottom to top approach divisive clusterin -- divide the cluster from bottom to top approach both we practice we worked
09-02-2026	pca , dbscan, intro nlp	pca (principal componenet anlaysis) -- reduce the high dimension to low dimension (dimension reduction technique) kernal function -- increase (low dimension to high dimension) dbscan -- clustering algo which require min 3 point to have core point & boundry point noise point which is outlier nlp we install framework -- nltk nlp -- nlu & nlg day by day text are increasing comments, chats nlu tokenization -- tokens words word_tokenize, sent_tokenize blankline_tokenize whitespace_tokenize (text cleanin)
10-02-2026	NLP Introduction	NLU - tokenization, stemming, lematization, stopwords, NER, POS NLG - wordcloud (text visualization) practices we worked ne_chunk is one functionality from NER
11-02-2026	NLP Algorithm Word embedding	bow -- bag of words -- convert text to number (words repet 2 times -- 2) tfidf -- tf * ldf (words are calucated to log caluculation -- decimal value) nlp -- text -- text clean -- as per the requirement -- text will convert number 0 & 1 then we imple all ml models nlp -- text -- word extraction -- ml clustering as well nltk library we understand
12-02-2026	spacy	spacy is more advanced then nltk one line code we complete all nlu technique load en_core_web_sm 326 stopwords spacy text summariozation token scoring sentence scoring heapq -- largest , nsmaallest, built the chatbot
13-02-2026	NLP	spacy introduce , text summarization, multilat translation text -- how to implement ml model we learned.. Regression, classification, cluster

16-02-2026	DEEP LEARNING -- ANN	Human neural network vs Artificial Neural network neurons, signal , nerural network, weight assignment, weight adjusted, input, hidden , outpuer singly layer nn vs multi layer nn forward prop weighs are assgined back proper weight adjusted new weight = old weigh - learnin * der loss/ der of old input layer activation function not reuqie hidden layer and output layer actiation function rquierd sigmoid activation funciton -- output layer, range -- 0 1 tanh apply output layer -- -1 to 1 relu activation we implement in hidden layer leeky relu we implement in -- hidden layer softmax activation function we implement in output layer range 0-1 prob function optimizer -- adam, adadelta, adagrade, adamax, rmsprop loss - mae, mse, log loss, classification -- binary cross entropy, category cross entropy, sparse category cross entropy install of tf, keras, torch, ultrayltics, medipipe, opencv langchain, langgraph & how to create env variable for your projects EPOC NEURAL NETWORK ARCHITECTURE
17-02-2026	Deep Learning - ANN	we built ANN model with single layer and 2 layer architecutre tf.keras.sequentiatl - establish the layer ann.add(neuron & densely connect, activate relu) ann.add (hidden layer) -- ann.add(sigmoid) as we worked on binary class classification optimizer -- adam, loss -- binary cross entropy, epoch , batchsize = 32 build ann model
18-02-2026	CNN - 1	Human vision vs computer vision human - see any image -- brain ever cerbral layer wise computer vision -- convoluion layer , max pool, flatten & fully connected layer image -- matrix (size will be more when image has more clarity matrix will define pixel size 0-255 cnn -- input matrix * filter = output matrix filter - feature detector output -- feature map convolution layer is math calculat filter will apply on input matrix to calculate otuput matrix bydefault strding is 1 s = 1 input matrix != output matrix padding concept -- p=1 (it iwll all 4 part to pad repalce with 0)
19-02-2026	CNN - 2	we have few images train test validation train & validation -- we create 2 folder happy face & sad faces testing -- pass few random images in both folder image shpas are happening how image are convered to array .flow_from_directory (train & test) rms prop optimizer loss function -- binarycross entropy activation relu, conv2d, maxpool2d, flatten & integrated to gradio application to detect
23-02-2026	keras pretreained model	resnet 50, resnet 50v2, enasnet, vgg16, vgg19 imagenet -- pretrained weights (dataset which converted to yaml file) data augumentation we are using image data generator we built how from 1image we created multipld generated images using help of keras image datagenerator
24-02-2026	open cv & haar cascade classifier	poc-1 : read, write, show image using opencv face detection using haar casscade classifier face & eye detection using opencv pedestistian detection car detection only video lines detection using opencv videocapture(0) -- immediaty it will open system webcam-- it will detect your face videocapture(videopath) -- it will detect video path video capture(1) -- inf any camera outside to the system then it will detect

25-02-2026	opencv practicles	color detection, ocr, video - frame cv2.videocapture, red mask, green mask
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